Rapid Weather Change Detection, Low Cloud Ceiling and Airport Visibility Cla

Authors / Researchers:
Jayawardhana H.M.S.I. (Student ID: IT22619976) | Specialization: Information Technology

EXECUTIVE ABSTRACT & SYSTEM SCOPE

Individual thesis addressing rapid onset fog, low stratus cloud ceilings, and visibility degradation at VCBI. Compares tree-based classifiers and gradient boosting to forecast instrument meteorological condition (IMC) threshold breaches.

1. Challenge of Rapid Weather Inversion

Tropical coastal airports experience rapid morning radiation fog and dense convection clouds that degrade visibility faster than standard 6-hour forecast cycles update.

Early warning systems are critical for diverting inbound flights before terminal airspace congestion occurs.

2. Classification Framework and Thresholds

Constructed multi-class targets based on ICAO visibility bands (CAT I, II, III minima and VFR/IFR delineations).

Evaluated Decision Tree (76.25%), Random Forest (76.31%), and optimized XGBoost classifier (78.15% overall accuracy).

3. Alert Integration & Flight-Time Windows

Integrated alert rules notify air traffic personnel when probability of visibility dropping below 3000m exceeds critical decision thresholds within the active 3-hour flight arrival window.